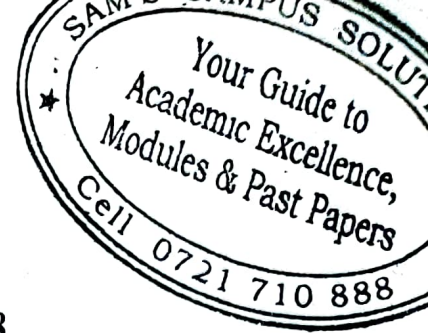




KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (MECHANICAL ENGINEERING)

EMM 307: ENGINEERING THERMODYNAMICS II

DATE: Friday, 23rd February, 2018

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS:

1. Answer any THREE questions
 2. All questions carry 20 marks
 3. Use neat and tidy diagrams
-
- ✓ 1. A steam power plant operates between a boiler pressure of 30 bar and condenser pressure of 0.04 bar. Calculate for these limits:-
 - i) The cycle efficiency
 - ii) The work ratio and
 - iii) The specific steam consumption
 - a) For a Carnot cycle using wet steam; [4marks]
 - b) For a Rankine cycle with dry saturated steam at entry to the turbine; [4marks]
 - c) For Rankine cycle in (b) when expansion process has an isentropic efficiency of 80% [4marks]
 - d) Neglecting feed pump work to do it for (c) when steam is superheated to 450°C. [4marks]
 - e) For a reheat cycle operated between the same pressure 30 bar and 0.040 bar. With a superheat temperature 450°C. [4marks]

The first expansion is carried out to the point where the steam is dry saturated and the steam is reheated to the original superheat temperature.

2. a) Show the thermal efficiency of the dual combustion cycle is given by:-

$$\eta = 1 - \frac{1}{r_v^{(\gamma-1)}} \left\{ \frac{\alpha \beta^\gamma - 1}{(\alpha - 1) + \alpha \gamma (\beta - 1)} \right\}$$

Where

$r_v = v_1/v_2$ = the adiabatic compression, volume ration

$\beta = v_4/v_3$ = cut-off ratio for constant pressure heat addition

$\alpha = p_3/p_2$ = constant volume heat addition pressure ration [8marks]

- b) An ideal dual combustion cycle has a volume ratio of the adiabatic compression of 15:1. At the beginning of the adiabatic compression the pressure, volume and temperature of the gas are 97KN/m², 0.084m³ and 28°C respectively. The maximum pressure and temperature of the cycle are 6.2 MN/m² and 1320°C, respectively.

For the cycle determine:

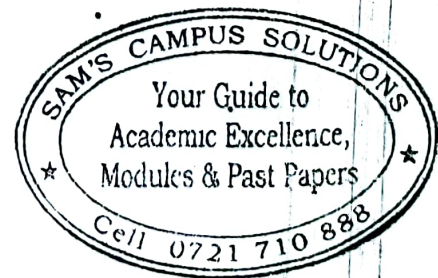
- i) The pressure, volume and temperature at the cycle process change points.
- ii) The net work done
- iii) The thermal efficiency
- iv) The heat received
- v) The work ratio
- vi) The mean effective pressure
- vii) The Carnot efficiency within the cycle temperature limits.

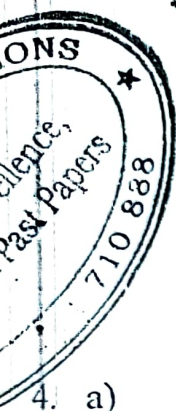
- i) Take $C_p = 1.005 \text{ KJ/kgK}$, $C_v = 0.717 \text{ KJ/kgK}$

[15marks]

3. The cycle of operations in a compression-ignition engine may be presented approximately as follows:

- i) Compression, according to $pV^{1.3} = \text{constant}$, through a volume compression ratio 14:1 from an initial pressure of 1 bar,
- ii) Constant volume combustion to a maximum pressure of 50 bar,
- iii) Constant pressure combustion during which the volume increases by 50%
- iv) Expansion, according to $pV^{1.3} = \text{constant}$, to the original volume,





v) Reduction of pressure at constant volume to the original pressure of 1 bar.

a) Sketch a P-V diagram showing these operations. [3marks]

b) Calculate the mean effective pressure. [9marks]

c) Determine the cylinder bore and the stroke of such an engine operating on the four stroke cycle for a brake power 1000KW at 3000rev/min if there are 4 cylinders and the piston stroke is equal to 1.25X cylinder bore diameter.

Assume the mechanical efficiency is 80% [8marks]

4. a) Draw a labeled diagram of the Orsat apparatus used for combustion analysis of gases, stating the order in which the product gases are absorbed in the apparatus. [5marks]

b) A petrol fuel has the following ultimate analysis:-C83.7%; H16.3% the dry product analysis by volume is:-CO₂ 11.8%; O₂ 3.7%; N₂ 84.5%

Determine, by Carbon-balance method, the air to fuel ratio used for combustion. [6marks]

i) Determine the stoichiometric air fuel ration for a petrol fuel approximating to hexane C₆ H₁₄.

ii) Deduce the chemical equation if the petrol is burnt in 20 per cent excess air and also

iii) The wet volumetric analysis of the product if all the water vapour is present. [9marks]

5. An open type gas turbine has two compressors in series. A high pressure turbine alone drives both compressors and a low-pressure turbine alone drives the external load. The pressure ration is the same for each compressor, the overall pressure ration for the engine being 1.6. The isentropic efficiency of each compressor is 0.85 and for each turbine 0.9. at inlet to the engine the air temperature pressure 20°C 1 bar respectively. The air leaving the low-pressure compressor is cooled at constant pressure at 20°C before entering the high pressure compressor. The gas temperature at entry to high pressure turbine is 800°C and on leaving this turbine the gas is reheated to 700°C before entering the low-pressure turbine. The exhaust from the low pressure turbine is used in heat exchanger to

raise the temperature of the air leaving the high pressure compressor by 75% of the maximum possible temperature rise.

a) Sketch i) the layout of the plant [2marks]

ii) a T-S diagram [2marks]

Showing the processes involved and marking on both diagrams the corresponding state points.

b) Determine:

i) The pressure of the air leaving the high pressure turbine. [4marks]

ii) The specific work output [4marks]

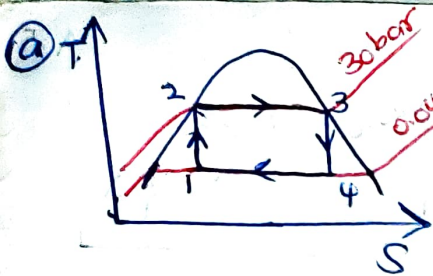
iii) The cycle efficiency and [4marks]

iv) The cycle efficiency of a simple air-standard Joule cycle having the same overall pressure ratio. [4marks]

NB. Assume the compressor and turbines operated adiabatically and that pressure losses in the combustion chamber, heat exchange and ducting are negligible.

For compression $C_p = 1.01 \text{ kJ/kgK}$ & $\gamma = 1.4$. For combustion and expansion

$C_p = 1.15 \text{ kJ/kgK}$ & $\gamma = 1.33$



$$h_2 = h_f \text{ at } 30 \text{ bar} = 1008 \text{ kJ/kg}$$

$$h_3 = h_g \text{ at } 30 \text{ bar} = 2803 \text{ kJ/kg}$$

$$\text{For } h_1, \text{ we need } S_1 = S_2 = 2.645 \text{ kJ/kg K (} S_f \text{ at } 30 \text{ bar)}$$

$$\text{For } h_4, \text{ we need } S_4 = S_3 = 6.186 \text{ kJ/kg K (} S_g \text{ at } 30 \text{ bar)}$$

$$S_1 = S_f + x_1 S_{fg} \Rightarrow 2.645 = 0.422 + 8.051 x_1$$

$$x_1 = 0.276$$

$$\therefore h_1 = h_f + x_1 h_{fg} = 121 + 0.276(2433) = 793 \text{ kJ/kg}$$

$$S_4 = S_f + x_4 S_{fg} \Rightarrow 6.186 = 0.422 + 8.051 x_4$$

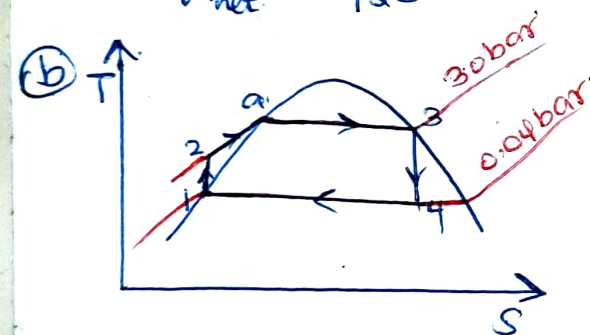
$$x_4 = 0.716$$

$$\therefore h_4 = h_f + x_4 h_{fg} = 121 + 0.716(2433) = 1863 \text{ kJ/kg}$$

$$W_i = h_2 - h_1 = 215 \text{ kJ/kg}; W_o = h_3 - h_4 = 940 \text{ kJ/kg}; Q_i = h_3 - h_2 = 1795 \text{ kJ/kg}$$

$$\eta = \frac{W_{net}}{Q_i} = \frac{940 - 215}{1795} = 0.404 \text{ or } 40.4\% \quad r_w = \frac{W_{net}}{W_o} = \frac{725}{940} = 0.771$$

$$SSC = \frac{3600}{W_{net}} = \frac{3600}{725} = 4.97 \text{ kg/kWh}$$



$$h_1 = 121 \text{ kJ/kg} \quad h_3 = 2803 \text{ kJ/kg} \quad h_4 = 1863 \text{ kJ/kg}$$

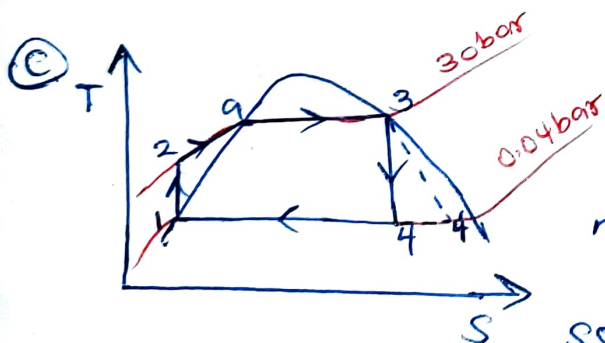
$$\text{Pump Work} = h_2 - h_1 = v(p_2 - p_1) \quad v = v_f \text{ at } 0.04 \text{ bar} = 0.001$$

$$= 0.001(30 - 0.04) \times \frac{10^5}{10^3} \text{ (Conversion of pressure from bar to kN/m}^2\text{)}$$

$$= 3 \text{ kJ/kg} \quad h_2 = h_1 + 3 = 124 \text{ kJ/kg}$$

$$W_i = 3 \text{ kJ/kg} \quad W_o = h_3 - h_4 = 940 \text{ kJ/kg} \quad Q_i = h_3 - h_2 = 2679$$

$$\eta = \frac{W_{net}}{Q_i} = \frac{940 - 3}{2679} = 0.35 \quad r_w = \frac{W_{net}}{W_o} = \frac{937}{940} = 0.997 \quad SSC = \frac{3600}{937} = 3.842 \text{ kg/kWh}$$



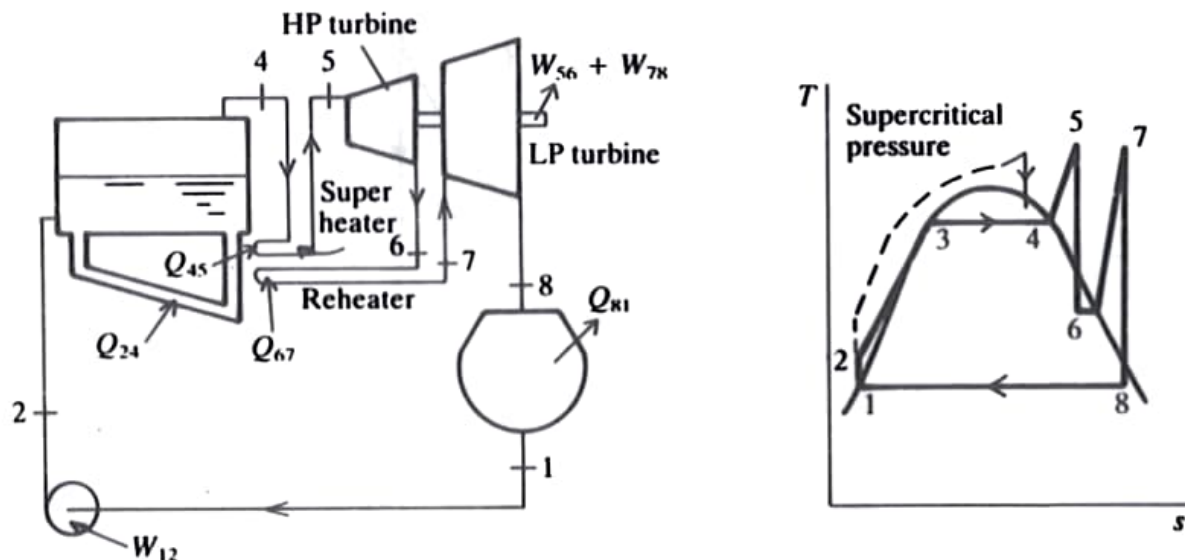
$$\eta_{iso} = \frac{h_3 - h'_4}{h_3 - h_4} = 0.8 \Rightarrow h_3 - h'_4 = 0.8(h_3 - h_4) = 752$$

$$W_i = 3 \text{ kJ/kg} \quad W_o = h_3 - h'_4 = 752 \quad Q_i = h_3 - h_2 = 2679$$

$$\eta = \frac{752 - 3}{2679} = 0.28 \quad r_w = \frac{749}{752} = 0.996$$

$$SSC = \frac{3600}{749} = 4.81 \text{ kg/kWh}$$

Fig. 11.12
The reheat cycle



11.4 Reheat cycle

With the reheat cycle the expansion takes place in two turbines. The steam expands in the high-pressure turbine to some intermediate pressure, and is then passed back to yet another bank of tubes in the boiler where it is reheated at constant pressure, usually to the original superheat temperature.* It then expands in the low-pressure turbine to the condenser pressure. The cycle appears as in Fig. 11.12, and the following example shows the effect of reheating on the ideal cycle efficiency and specific consumption. A boiler pressure of 30 bar has been assumed for the sake of comparison with the other examples in this chapter, although this is well below the most economical pressure for which to design a reheat cycle plant.

Example 11.4 Find the ideal cycle efficiency and specific steam consumption of a reheat cycle (Fig. 11.12) operating between pressures of 30 and 0.04 bar, with a superheat temperature of 450°C . Assume that the first expansion is carried out to the point where the steam is dry saturated and that the steam is reheated to the original superheat temperature. The feed pump term may be neglected.

From tables,

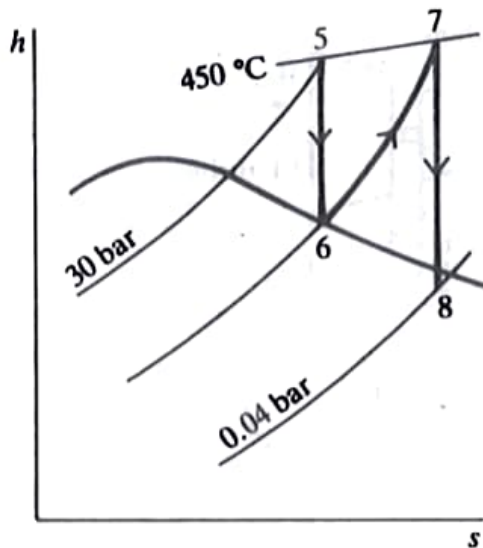
$$h_2 \approx h_1 = 121 \text{ kJ/kg}, \quad h_5 = 3343 \text{ kJ/kg},$$

$$s_5 = 7.082 \text{ kJ/kg K}$$

To find the intermediate reheat pressure p_6 , we find from the saturation table the pressure at which $s_g = s_5$. This gives

$$p_6 = p_7 = 2.3 \text{ bar} \quad \text{and hence} \quad h_6 = 2713 \text{ kJ/kg}$$

Fig. 11.13
Use of the h - s
diagram



From the superheat table,

$$h_7 = 3381 \text{ kJ/kg} \quad \text{and} \quad s_7 = 8.310 \text{ kJ/kg K}$$

At the turbine outlet,

$$x_8 = 0.980 \quad \text{and} \quad h_8 = 2505 \text{ kJ/kg}$$

The total quantity of heat transferred to the steam in the boiler is

$$Q_{25} + Q_{67} = (h_5 - h_2) + (h_7 - h_6) = 3222 + 668 = 3890 \text{ kJ/kg}$$

and the total turbine work is

$$\begin{aligned} W_{56} + W_{78} &= (h_6 - h_5) + (h_8 - h_7) \\ &= -630 - 876 = -1506 \text{ kJ/kg} \end{aligned}$$

The cycle efficiency is therefore

$$\eta = \frac{|(W_{56} + W_{78})|}{Q_{25} + Q_{67}} = \frac{1506}{3890} = 0.387$$

and the specific steam consumption is

$$SSC = \frac{1}{|W|} = \frac{3600}{1506} = 2.39 \text{ kg/kW h}$$

- (c) For the constant pressure process.

$$\text{Stroke} = 16 - 1 = 15 \text{ volumes}$$

$$\therefore 3\% \text{ of stroke} = 15 \times 0.03 = 0.45 \text{ volumes}$$

$$\therefore \text{Volume at the end of the constant pressure heat addition}$$

$$= 1 + 0.45 = 1.45 \text{ volumes} = V_4$$

$$\frac{V_4}{T_4} = \frac{V_3}{T_3}$$

$$\therefore T_4 = T_3 \frac{V_4}{V_3} = 1234 \times \frac{1.45}{1} = 1789 \text{ K}$$

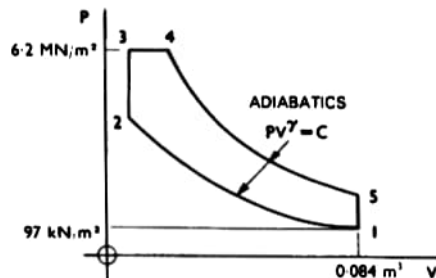
$$t_4 = 1789 - 273 = 1516^\circ\text{C}$$

EXAMPLE 130

An ideal dual combustion cycle has a volume ratio of the adiabatic compression of 15:1. At the beginning of the adiabatic compression the pressure, volume and temperature of the gas are 97 kN/m^2 , 0.084 m^3 and 28°C , respectively. The maximum pressure and temperature of the cycle are 6.2 MN/m^2 and 1320°C , respectively. For the cycle, determine,

- the pressure, volume and temperature at the cycle process change points,
- the net work done,
- the thermal efficiency,
- the heat received,
- the work ratio,
- the mean effective pressure,
- the Carnot efficiency within the cycle temperature limits.

Take, $c_p = 1.005 \text{ kJ/kg K}$, $c_v = 0.717 \text{ kJ/kg K}$.



$$(a) P_1 = 97 \text{ kN/m}^2, V_1 = 0.084 \text{ m}^3, t_1 = 28^\circ\text{C}$$

$$V_2 = \frac{0.084}{15} = 0.0056 \text{ m}^3$$

$$c_p/c_v = \gamma = \frac{1.005}{0.717} = 1.4$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\therefore P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma = 97 \times 15^{1.4} = 97 \times 44.3 = 4297 \text{ kN/m}^2$$

$$= 4.297 \text{ MN/m}^2$$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1} \right)^{\gamma-1} \quad \text{and} \quad T_1 = 28 + 273 = 301 \text{ K}$$

$$\therefore T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 301 \times 15^{0.4} = 301 \times 2.954 = 889 \text{ K}$$

$$t_2 = 889 - 273 = 616^\circ\text{C}$$

$$\frac{P_3}{T_3} = \frac{P_2}{T_2}$$

$$\therefore T_3 = T_2 \frac{P_3}{P_2} = 889 \times \frac{6.2}{4.297} = 1283 \text{ K}$$

$$t_3 = 1283 - 273 = 1010^\circ\text{C}$$

$$V_3 = V_2 = 0.0056 \text{ m}^3$$

$$\frac{V_4}{T_4} = \frac{V_3}{T_3} \quad \text{and} \quad T_4 = 1320 + 273 = 1593 \text{ K}$$

$$\therefore V_4 = V_3 \frac{T_4}{T_3} = 0.0056 \times \frac{1593}{1283} = 0.00695 \text{ m}^3$$

$$P_4 = P_3 = 6.2 \text{ MN/m}^2 \quad V_5 = V_1 = 0.084 \text{ m}^3$$

$$P_4 V_4^\gamma = P_5 V_5^\gamma$$

$$\therefore P_5 = P_4 \left(\frac{V_4}{V_5} \right)^\gamma = 6.2 \times \left(\frac{0.00695}{0.084} \right)^{1.4} = \frac{6.2}{12.09^{1.4}} = \frac{6.2}{32.8}$$

$$= 0.189 \text{ MN/m}^2 = 189 \text{ kN/m}^2$$

$$\frac{T_5}{T_4} = \left(\frac{V_4}{V_5} \right)^{\gamma-1}$$

$$\therefore T_3 = T_4 \left(\frac{V_4}{V_3} \right)^{(\gamma-1)} = 1593 \times \left(\frac{0.00695}{0.084} \right)^{0.4} = \frac{1593}{12.09^{0.4}} = \frac{1593}{2.71}$$

$$= 588 \text{ K}$$

$$t_3 = 588 - 273 = 315^\circ\text{C}$$

Hence,

$$\begin{aligned} P_1 &= 97 \text{ kN/m}^2 & V_1 &= 0.084 \text{ m}^3 & t_1 &= 28^\circ\text{C} \\ P_2 &= 4.297 \text{ MN/m}^2 & V_2 &= 0.0056 \text{ m}^3 & t_2 &= 616^\circ\text{C} \\ P_3 &= 6.2 \text{ MN/m}^2 & V_3 &= 0.0056 \text{ m}^3 & t_3 &= 1010^\circ\text{C} \\ P_4 &= 6.2 \text{ MN/m}^2 & V_4 &= 0.00695 \text{ m}^3 & t_4 &= 1320^\circ\text{C} \\ P_5 &= 189 \text{ kN/m}^2 & V_5 &= 0.084 \text{ m}^3 & t_5 &= 315^\circ\text{C} \end{aligned}$$

$$\begin{aligned} \text{(b) } \oint W &= P_3(V_4 - V_3) + \frac{\{(P_4 V_4 - P_3 V_3) - (P_2 V_2 - P_1 V_1)\}}{(\gamma - 1)} \\ &= 6200(0.00695 - 0.0056) \\ &\quad + \frac{\{[(6200 \times 0.00695) - (189 \times 0.084)] - [(4297 \times 0.0056) - (97 \times 0.084)]\}}{(1.4 - 1)} \\ &= (6200 \times 0.00135) + \frac{\{(43.1 - 15.9) - (24.1 - 8.15)\}}{0.4} \\ &= 8.37 + \frac{(27.2 - 15.95)}{0.4} = 8.37 + \frac{11.25}{0.4} = 8.37 + 28.13 \\ &= 36.5 \text{ kJ} \end{aligned}$$

$$\begin{aligned} \text{(c) Thermal } \eta &= 1 - \frac{(T_3 - T_1)}{(T_3 - T_2) + \gamma(T_4 - T_3)} \\ &= 1 - \frac{(588 - 301)}{(1283 - 889) + 1.4(1593 - 1283)} \\ &= 1 - \frac{287}{394 + (1.4 \times 310)} = 1 - \frac{287}{394 + 434} \\ &= 1 - \frac{287}{828} = 1 - 0.347 \\ &= 0.653 \\ &= 0.653 \times 100 \\ &= 65.3\% \end{aligned}$$

Alternatively,

$$\text{Thermal } \eta = 1 - \frac{1}{r_v^{(\gamma-1)}} \cdot \frac{(\alpha\beta^\gamma - 1)}{(\alpha + 1) + \alpha\gamma(\beta - 1)}$$

and

$$r_v = 15$$

$$\alpha = \frac{P_3}{P_2} = \frac{6.2}{4.297} = 1.443$$

$$\beta = \frac{V_4}{V_3} = \frac{0.00695}{0.0056} = 1.241$$

$$\begin{aligned} \therefore \text{Thermal } \eta &= 1 - \frac{1}{15^{0.4}} \times \frac{\{(1.443 \times 1.241^{1.4}) - 1\}}{(1.443 - 1) + (1.443 \times 1.4)(1.241 - 1)} \\ &= 1 - \frac{1}{2.95} \times \frac{\{(1.443 \times 1.353) - 1\}}{0.443 + (2.02 \times 0.241)} \\ &= 1 - \frac{1}{2.95} \times \frac{(1.952 - 1)}{0.448 + 0.487} \\ &= 1 - \frac{1}{2.95} \times \frac{0.952}{0.93} = 1 - 0.347 \\ &= 0.653 \end{aligned}$$

$$\text{(d) } \oint W = \text{Heat received} \times \text{Thermal } \eta$$

$$\begin{aligned} \therefore \text{Heat received} &= \frac{\oint W}{\text{Thermal } \eta} \\ &= \frac{36.5}{0.653} = 55.9 \text{ kJ} \end{aligned}$$

Alternatively,

$$R = c_p - c_v = 1.005 - 0.717 = 0.288 \text{ kJ/kg K}$$

$$m = \frac{P_1 V_1}{RT_1} = \frac{97 \times 0.084}{0.288 \times 301} = 0.095 \text{ kg}$$

$$\begin{aligned} \text{Heat received} &= mc_v(T_3 - T_2) + mc_p(T_4 - T_3) \\ &= \{0.094 \times 0.717 \times (1283 - 889)\} \\ &\quad + \{0.094 \times 1.006 \times (1593 - 1283)\} \\ &= (0.094 \times 0.717 \times 394) + (0.094 \times 1.006 \times 310) \end{aligned}$$

$$= 26.6 + 29.3$$

$$= \underline{55.9 \text{ kJ}}$$

$$\begin{aligned} \text{(e) Work ratio} &= \frac{\oint W}{\text{Positive work done}} \\ &= \frac{36.5}{P_3(V_4 - V_3) + \frac{(P_4 V_4 - P_5 V_5)}{(\gamma - 1)}} \\ &= \frac{36.5}{8.37 + \frac{27.2}{0.4}} \\ &= \frac{36.5}{837 + 68} \\ &= \frac{36.5}{76.37} \\ &= \underline{0.478} \end{aligned}$$

$$\begin{aligned} \text{(f) Mean effective pressure} &= \frac{\oint W}{(V_1 - V_2)} \\ &= \frac{36.5}{(0.084 - 0.0056)} \\ &= \frac{36.5}{0.0784} \\ &= \underline{465.6 \text{ kN/m}^2} \end{aligned}$$

$$\begin{aligned} \text{(g) Carnot } \eta &= \frac{T_4 - T_1}{T_4} \\ &= \frac{1593 - 301}{1593} \\ &= \frac{1292}{1593} \\ &= \underline{0.81} \\ &= 0.81 \times 100 \\ &= \underline{81\%} \end{aligned}$$

The reciprocating piston engine is a device in which the steady flow energy equation can be applied across the whole unit from inlet to outlet at points where the fluid states can be said to be uniform.

However, the various processes occurring within the device are either unsteady in the case of suction and delivery or non-flow in compression, combustion and expansion.

The choice of energy equation is therefore a matter of careful thought and is an illuminating exercise.

There are several efficiencies relevant to this topic and they need careful qualification.

For example

$$\text{Mechanical efficiency} = \frac{\text{Brake work transfer}}{\text{Indicated work}}$$

$$\text{Volumetric efficiency} = \frac{\text{Induced volume}}{\text{Swept volume}}$$

$$\text{Thermal efficiency} = \frac{\text{Work transfer}}{\text{Energy input}}$$

to mention but three.

1. The cycle of operations in a compression-ignition engine may be represented approximately as follows:

- (i) Compression, according to $pv^{1.3} = \text{constant}$, through a volume compression ratio of 14:1 from an initial pressure of 1 bar,
- (ii) Constant volume combustion to a maximum pressure of 50 bar,
- (iii) Constant pressure combustion during which the volume increases by 50%,
- (iv) Expansion, according to $pv^{1.3} = \text{constant}$, to the original volume,
- (v) Reduction of pressure at constant volume to the original pressure of 1 bar.

Sketch a p - v diagram showing these operations and calculate

- (a) the mean effective pressure,
- (b) the cylinder bore and stroke of such an engine operating on the four stroke cycle for a brake power of 100 kW at 3000 rev/min if there are 4 cylinders and the piston stroke is equal to $1.25 \times$ cylinder bore diameter.

Assume that the mechanical efficiency is 80%. (Aston University 1975)

Figure 2.1 refers.

The mean effective pressure (mep) is given by

$$\text{mep} = \frac{\text{net spec.work}}{\text{swept spec.volume}} = \frac{1W_2 + 3W_4 + 4W_5}{v_1 - v_2} \quad (2W_3 = 5W_1 = 0)$$

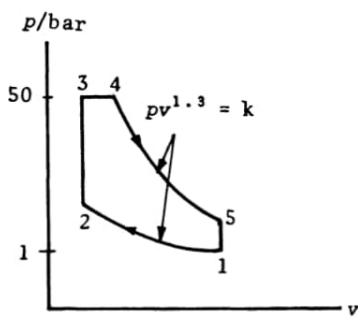


Figure 2.1

$$\begin{aligned} \text{mep} &= \frac{\frac{R(T_1 - T_2)}{n-1} + R(T_4 - T_3) + \frac{R(T_4 - T_5)}{n-1}}{v_1 \left(1 - \frac{v_2}{v_1}\right)} \\ &= p_1 \frac{\frac{(T_1 - T_2)}{n-1} + (T_4 - T_3) + \frac{(T_4 - T_5)}{n-1}}{T_1 \left(1 - \frac{1}{r}\right)} \end{aligned}$$

$$T_2 = T_1 \left(\frac{v_1}{v_2}\right)^{n-1} = T_1 (14)^{0.3} = 2.207 T_1$$

$$\begin{aligned} T_3 &= T_2 \frac{p_3}{p_2} = T_2 \frac{p_3 p_1}{p_1 p_2} = T_2 \frac{p_3}{p_1} \left(\frac{v_2}{v_1}\right)^n = 2.207 T_1 \left[50 \left(\frac{1}{14}\right)^{1.3}\right] \\ &= 3.571 T_1 \end{aligned}$$

$$T_4 = T_3 \frac{v_4}{v_3} = 1.5 T_3 = 5.357 T_1$$

$$\begin{aligned} T_5 &= T_4 \left(\frac{v_4}{v_5}\right)^{n-1} = T_4 \left(\frac{v_4}{v_1}\right)^{n-1} = 5.357 T_1 \left(\frac{1.5 v_2}{v_1}\right)^{0.3} \\ &= 5.357 T_1 \times \left(\frac{1.5}{14}\right)^{0.3} = 2.741 T_1 \end{aligned}$$

$$\begin{aligned} \text{Thus mep} &= p_1 \frac{\frac{T_1(1 - 2.207)}{0.3} + T_1(5.357 - 3.571) + \frac{T_1(5.357 - 2.741)}{0.3}}{T_1 \left(1 - \frac{1}{14}\right)} \\ &= 1 \text{ bar} \frac{14}{13} (-4.023 + 1.786 + 8.720) = 6.98 \text{ bar} \end{aligned}$$

The equivalent mean effective pressure on the brake basis is given by

$$\text{bmep} = 0.8 \text{mep} = 0.8 \times 6.98 \text{ bar} = 5.59 \text{ bar} = p_b$$

$$\text{Swept volume rate} = \frac{\dot{W}}{p_b} = \frac{100 \frac{\text{kJ}}{\text{s}}}{559 \frac{\text{kN}}{\text{m}^2}} = 0.1789 \frac{\text{m}^3}{\text{s}}$$

$$\text{Thus } 0.1789 \frac{\text{m}^3}{\text{s}} = 4 \times \frac{N}{2} \times \frac{\pi D^2}{4} \times 1.25 D$$

where N = speed in rev/min and D = bore in m (4 stroke cycle)

$$D^3 = \frac{0.1789 \times 2 \frac{\text{m}^3}{\text{s}}}{3000 \frac{\text{rev}}{\text{min}} \left[\frac{\text{min}}{60 \text{ s}}\right] \times \pi \times 1.25} = 1.822 \times 10^{-3} \text{ m}^3$$

$$D = 0.122 \text{ m} = \text{bore}$$

$$\text{and stroke} = 1.25 \times 0.122 = 0.153 \text{ m}$$

Example 15.4 A petrol has the following ultimate analysis:

C 83.7%; H 16.3%

If the dry products analysis by volume is

CO₂ 11.8%; O₂ 3.7%; N₂ 84.5%

determine the air/fuel ratio used.

Carbon balance method

Treating the volume fractions as mole fractions, 1 kmol of dry products contains 0.845 kmol of N₂. Now 0.845 kmol of N₂ are brought in with

$$0.845 \times 28 \left(\frac{100}{76.7} \right) = 30.85 \text{ kg of air}$$

1 kmol of dry products contains 0.118 kmol of CO₂, which in turn contain 0.118 × 12 kg of carbon. This amount of carbon implies the combustion of

$$\frac{0.118 \times 12}{0.837} = 1.692 \text{ kg of petrol}$$

Thus 1 kmol of dry products is produced when 30.85 kg of air are burnt with 1.692 kg of petrol, i.e.

$$\text{air/fuel ratio} = \frac{30.85}{1.692} = 18.2$$

The foregoing method, using a carbon balance between reactants and products, is the most satisfactory when carbon is the predominant element in the fuel. The result may be inaccurate, however, if the exhaust gases are smoky, indicating unburnt carbon not included in the products analysis. An oxygen–hydrogen balance can be made as a check, although since the relevant quantities are relatively small the result is more open to error from inaccuracies of measurement.

Oxygen–hydrogen balance method

First, 1 kmol of dry products contains

$$32(0.118 + 0.037) = 4.960 \text{ kg of O}_2$$

But the oxygen accompanying the nitrogen is

$$0.845 \times 28 \left(\frac{23.3}{76.7} \right) = 7.187 \text{ kg}$$

Hence (7.187 – 4.960) = 2.227 kg of O₂ have combined with the hydrogen in the fuel to form water vapour. It follows from equation (15.7) that 2.227/8 kg of H₂ are involved, which corresponds to

$$\frac{2.227}{8 \times 0.163} = 1.708 \text{ kg of petrol}$$

331

Therefore 30.85 kg of air are associated with 1.708 kg of fuel, i.e.

$$\text{air/fuel ratio} = \frac{30.85}{1.708} = 18.1$$

Example 15.1

Determine the stoichiometric air/fuel ratio for a petrol approximating to hexane C_6H_{14} . Hence deduce the chemical equation if the petrol is burnt in 20 per cent excess air, and the wet volumetric analysis of the products (a) if all the water vapour is present, and (b) if the products are cooled to an atmospheric pressure and temperature of 1 bar and 15°C . Determine also the dry volumetric analysis.

Finally, estimate the chemical equation if only 80 per cent of the air required for stoichiometric combustion is provided.

Chemical equation

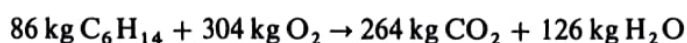
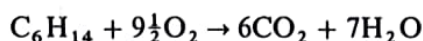
Since the combustion is theoretically complete with a stoichiometric mixture, the products will consist only of CO_2 , H_2O and the N_2 brought in with the air. The amount of oxygen required can be found by first balancing the carbon

* The chief exception is the rocket motor, described in section 18.4.2. In this case the necessary oxygen is obtained from liquid oxygen, liquid ozone, or some oxygen carrier such as hydrogen peroxide or nitric acid.

325

Applications to Particular Fluids**[15.2]**

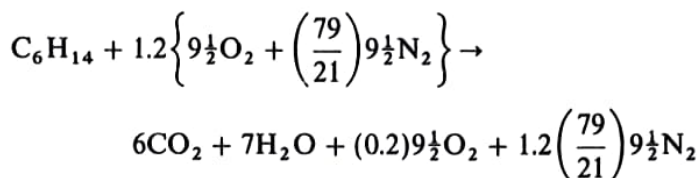
and hydrogen atoms on each side of the equation and then summing the number of atoms of oxygen in the products:



Since air contains 23.3 per cent by mass of oxygen, we have

$$\text{stoichiometric air/fuel ratio} = \frac{304 \times 100}{86 \times 23.3} = 15.17$$

If there is 20 per cent excess air, the equation, including the nitrogen, becomes

*Wet volumetric analysis*

(a) The amount-of-substance of the products is

$$6(CO_2) + 7(H_2O) + 1.9(O_2) + 42.89(N_2) = 57.79 \text{ kmol}$$

Now the mole fraction of a constituent equals the volume fraction (equation (14.18)). The wet volumetric analysis if all the water vapour is present is therefore

$$CO_2 \quad \frac{6 \times 100}{57.79} = 10.38\%; \quad H_2O \quad \frac{7 \times 100}{57.79} = 12.11\%$$

$$O_2 \quad \frac{1.9 \times 100}{57.79} = 3.29\%; \quad N_2 \quad \frac{42.89 \times 100}{57.79} = 74.22\%$$

4. An open-type gas turbine has two compressors in series. A high-pressure turbine alone drives both compressors and a low-pressure turbine alone drives the external load. The pressure ratio is the same for each compressor, the overall pressure ratio for the engine being 16. The isentropic efficiency of each compressor is 0.85 and for each turbine it is 0.9. At inlet to the engine the air temperature and pressure are 20 °C and 1.0 bar. The air leaving the low-pressure compressor is cooled at constant pressure to 20 °C before entering the high-pressure compressor. The gas temperature at entry to the high-pressure turbine is 800 °C and on leaving this turbine the gas is reheated to 700 °C before entering the low-pressure turbine. The exhaust from the low-pressure turbine is used in a heat exchanger to raise the temperature of the air leaving the high-pressure compressor by 75% of the maximum possible temperature rise.

Sketch the layout of the plant and show on a temperature-entropy diagram the processes involved marking on both diagrams the corresponding points.

Determine:

- (a) the pressure of the gas leaving the high-pressure turbine,
- (b) the specific work output,
- (c) the cycle efficiency and
- (d) the cycle efficiency of the simple air-standard (Joule) cycle having the same overall pressure ratio.

Assume that the compressors and turbines operate adiabatically and that pressure losses in the combustion chamber, heat exchangers and ducting are negligible. For compression take $c_p = 1.01 \text{ kJ/(kg K)}$ and $\gamma = 1.4$. For combustion and expansion take $c_p = 1.15 \text{ kJ/(kg K)}$ and $\gamma = 1.33$.
(London University 1974 modified)

Figure 7.4 refers

$$T_{2s} = T_1 \left[\frac{p_2}{p_1} \right]^{k_c} = 293 \text{ K} (4)^{0.286} = 435.6 \text{ K}$$

$$T_{2s} - T_1 = 142.6 \text{ K}$$

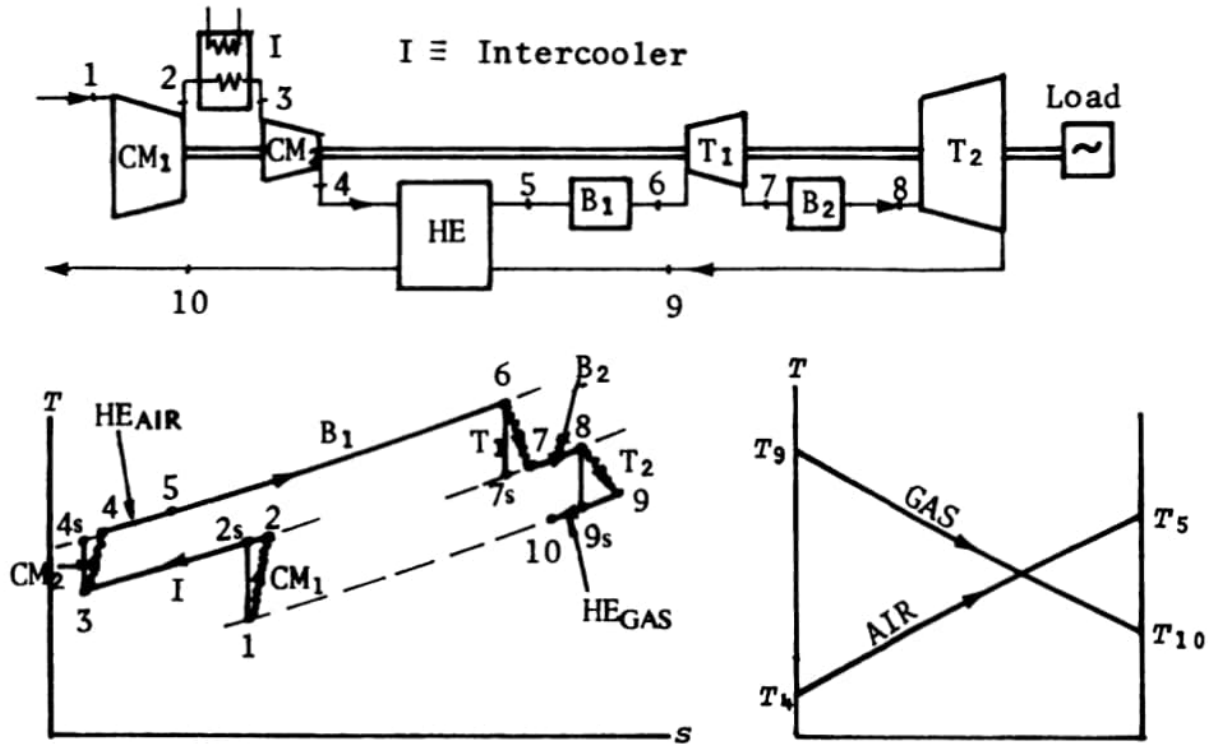


Figure 7.4

Distance Along HE

$$T_2 - T_1 = \frac{142.6}{0.85} = 167.7 \text{ K}$$

$$T_2 = 460.7 \text{ K}$$

Now $T_4 - T_3 = 167.7 \text{ K}$; $T_4 = 460.7 \text{ K}$ (since $\frac{p_4}{p_3} = \frac{p_2}{p_1}$ and $T_3 = T_1$)

and $W_{T1} = W_{C1} + W_{C2}$

or $T_6 - T_7 = \frac{c_{pC}}{c_{pT}}(T_4 - T_3 + T_2 - T_1) = \frac{1.01}{1.15}(2 \times 167.7) \text{ K} = 294.6 \text{ K}$

$$T_7 = T_6 - 294.6 \text{ K} = 1073 - 294.6 = 778.4 \text{ K}$$

$$T_6 - T_{7s} = \frac{294.6}{0.9} = 327.3 \text{ K}$$

$$T_{7s} = 1073 - 327.3 = 745.7 \text{ K}$$

$$\frac{p_7}{p_6} = \left[\frac{T_{7s}}{T_6} \right]^{1/k_T} = \left[\frac{745.7}{1073} \right]^{4.03} = 0.231$$

$$p_7 = 16 \times 0.231 = 3.69 \text{ bar}$$

$$T_{9s} = T_8 \left[\frac{p_9}{p_8} \right]^{k_T} = 973 \text{ K} \left[\frac{1}{3.69} \right]^{0.248} = 703.8 \text{ K}$$

$$T_8 - T_{9s} = 269.2 \text{ K}$$

$$T_8 - T_9 = 0.9 \times 269.2 = 242.3 \text{ K}$$

$$T_9 = 730.7 \text{ K}$$

In the heat exchanger HE

$$T_5 = T_4 + 0.75(T_9 - T_4) = 460.7 + 0.75(730.7 - 460.7) = 663.2 \text{ K}$$

and $c_{pT}(T_9 - T_{10}) = c_{pC}(T_5 - T_4)$

$$\text{or } T_9 - T_{10} = \frac{1.01}{1.15}(663.2 - 460.7) = 177.9 \text{ K}$$

$$T_{10} = 730.7 - 177.9 = 552.8 \text{ K}$$

$$\eta = \frac{w_{\text{net}}}{q_{+\text{ve}}} = \frac{c_{pT}(T_8 - T_9)}{c_{pT}[(T_6 - T_5) + (T_8 - T_7)]}$$

$$= \frac{242.3}{(1073 - 663.2 + 973 - 778.4)} = 0.401$$

and the specific work output is given by

$$w_{\text{net}} = 1.15 \times 242.3 = 278.6 \text{ kJ/kg}$$

Joule cycle

$$T_B = T_A \left[\frac{p_B}{p_A} \right]^{k_C} = 293(16)^{0.286} = 647.5 \text{ K}$$

$$T_C = 1073 \text{ K}$$

$$T_D = 1073 \left[\frac{1}{16} \right]^{0.248} = 539.5 \text{ K}$$

$$\eta_{\text{Joule}} = \frac{c_{pT}(\Delta T_T) - c_{pC}(\Delta T_C)}{c_{pT}\Delta T_B}$$

$$= \frac{1.15(1073 - 539.5) - 1.01(647.5 - 293)}{1.15(1073 - 647.5)} = 0.522$$



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
MECHANICAL ENGINEERING
EMM 307: ENGINEERING THERMODYNAMICS II

3RD TRIMESTER 2021/2022

TIME: 2 HOURS

-
- 1. Answer Question ONE (Compulsory). (30 marks)**
 - 2. Choose any other TWO Questions. (20 marks each)**
 - 3. Use, neat, clear and tidy sketches.**
 - 4. Steam tables and h-s chart provided. Usual notation used.**

Q1. A steam power plant operates between a boiler pressure of 40 bar and a condenser pressure of 0.035 bar. Calculate for these limits:-

- (i) The cycle efficiency
 - (ii) The work ratio and
 - (iii) The specific steam consumption
- (a) For a Carnot cycle using wet steam; [6 marks]
 - (b) For a Rankine cycle with dry saturated steam at entry to the turbine; [6 marks]
 - (c) For Rankine cycle in (b) when the expansion process has an isentropic efficiency of 80%. [6 marks]
 - (d) Neglecting feed pump work do it for (c) when steam is superheated to 450⁰ C. [6 marks]
 - (e) For a reheat cycle operated between the same pressures 40 bar and 0.035 bar. With a superheat temperature of 450⁰ C, [6 marks]

The first expansion is carried out to the point where the steam is dry saturated and the steam is reheated to the original temperature.

Example 8.1

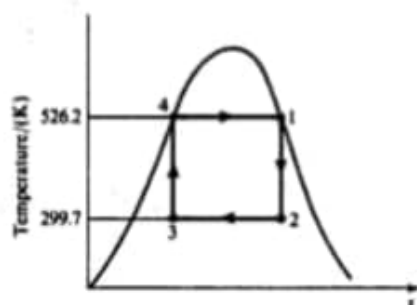
A steam power plant operates between a boiler pressure of 42 bar and a condenser pressure of 0.035 bar. Calculate for these limits the cycle efficiency, the work ratio, and the specific steam consumption:

Not exact
question

- for a Carnot cycle using wet steam;
- for a Rankine cycle with dry saturated steam at entry to the turbine;
- for the Rankine cycle of (ii), when the expansion process has an isentropic efficiency of 80%.

Solution (i) A Carnot cycle is shown in Fig. 8.5.

5 Carnot cycle
Example 8.1(a)



$$T_1 = \text{saturation temperature at 42 bar} \\ = 253.2 + 273 = 526.2 \text{ K}$$

$$T_2 = \text{saturation temperature at 0.035 bar} \\ = 26.7 + 273 = 299.7 \text{ K}$$

Then from equation (5.1)

$$\eta_{\text{Carnot}} = \frac{T_1 - T_2}{T_1} = \frac{526.2 - 299.7}{526.2} = 0.432 \quad \text{or} \quad 43.2\%$$

259

Urheberrechtlich geschütztes Material

Cycles

Also Heat supplied $= h_1 - h_4 = h_{fg}$ at 42 bar $= 1698 \text{ kJ/kg}$

$$\text{Then } \eta_{\text{Carnot}} = \frac{\text{Net work output, } -\sum W}{\text{Gross heat supplied}} = 0.432$$

Therefore $-\sum W = 0.432 \times 1698$,

i.e. Net work output, $-\sum W = 734 \text{ kJ/kg}$

To find the gross work of the expansion process it is necessary to calculate h_2 , using the fact that $s_1 = s_2$.

From tables

$$h_1 = 2800 \text{ kJ/kg} \quad \text{and} \quad s_1 = s_2 = 6.049 \text{ kJ/kg K}$$

Using equation (4.10)

$$s_2 = 6.049 = s_{f1} + x_2 s_{fg1} = 0.391 + x_2 8.13$$

therefore

$$x_2 = \frac{6.049 - 0.391}{8.13} = 0.696$$

Then using equation (2.2)

$$h_2 = h_{f1} + x_2 h_{fg1} = 112 + (0.696 \times 2438) = 1808 \text{ kJ/kg}$$

Hence, from equation (8.2)

$$-W_{12} = h_1 - h_2 = 2800 - 1808 = 992 \text{ kJ/kg}$$

Therefore we have, using equation (8.13),

$$\text{Work ratio} = \frac{\text{net work output}}{\text{gross work output}} = \frac{734}{992} = 0.739$$

Using equation (8.14)

$$ssc = \frac{1}{734}$$

i.e. $ssc = 0.00136 \text{ kg/kW s}$
 $= 4.91 \text{ kg/kW h}$

(ii) The Rankine cycle is shown in Fig. 8.6.
 As in part (i)

$$h_1 = 2800 \text{ kJ/kg and } h_2 = 1808 \text{ kJ/kg}$$

Also, $h_3 = h_f$ at 0.035 bar = 112 kJ/kg

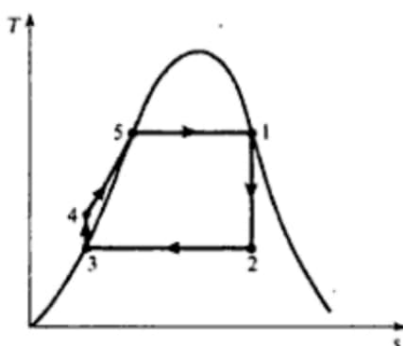
Using equation (8.10), with $v = v_f$ at 0.035 bar

$$\begin{aligned} \text{Pump work} &= v_f(p_4 - p_3) = 0.001 \times (42 - 0.035) \times \frac{10^5}{10^3} \\ &= 4.2 \text{ kJ/kg} \end{aligned}$$

Urheberrechtlich geschütztes Material

8.1 The Rankine cycle

diagram
 1(b)



Using equation (8.2)

$$-W_{12} = h_1 - h_2 = 2800 - 1808 = 992 \text{ kJ/kg}$$

Then using equation (8.8)

$$\eta_R = \frac{(h_1 - h_2) - (h_4 - h_3)}{(h_1 - h_3) - (h_4 - h_3)} = \frac{992 - 4.2}{(2800 - 112) - 4.2} = 0.368$$

i.e. $\eta_R = 36.8\%$

Using equation (8.13)

$$\text{Work ratio} = \frac{\text{net work output}}{\text{gross work output}} = \frac{992 - 4.2}{992} = 0.996$$

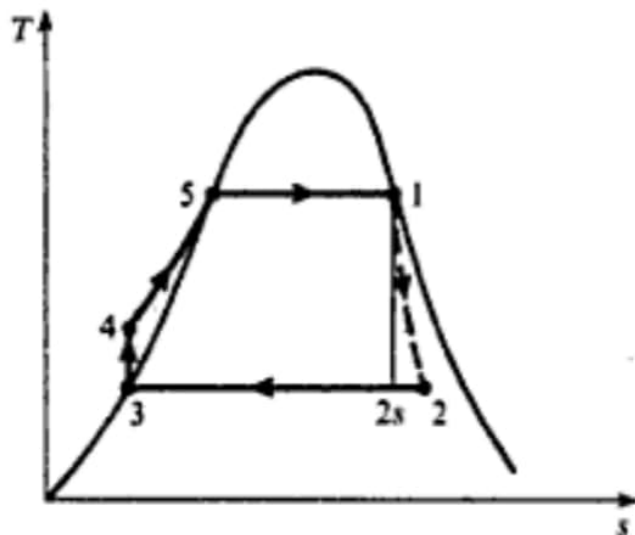
Using equation (8.14)

$$ssc = \frac{1}{-\sum W}$$

i.e. $ssc = \frac{1}{(992 - 4.2)} = 0.00101 \text{ kg/kW s} = 3.64 \text{ kg/kW h}$

(iii) The cycle with an irreversible expansion process is shown in Fig. 8.7.
 Using equation (8.12)

$$\text{Isentropic efficiency} = \frac{h_1 - h_2}{h_1 - h_{2s}} = \frac{-W_{12}}{-W_{12s}}$$



Urheberrechtlich geschützt

therefore

$$0.8 = \frac{-W_{12}}{992}$$

i.e. $-W_{12} = 0.8 \times 992 = 793.6 \text{ kJ/kg}$

Then the cycle efficiency is given by

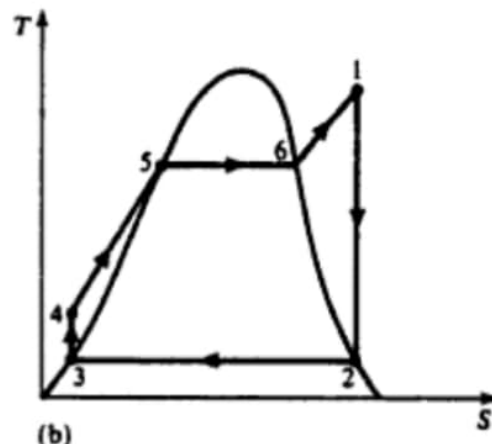
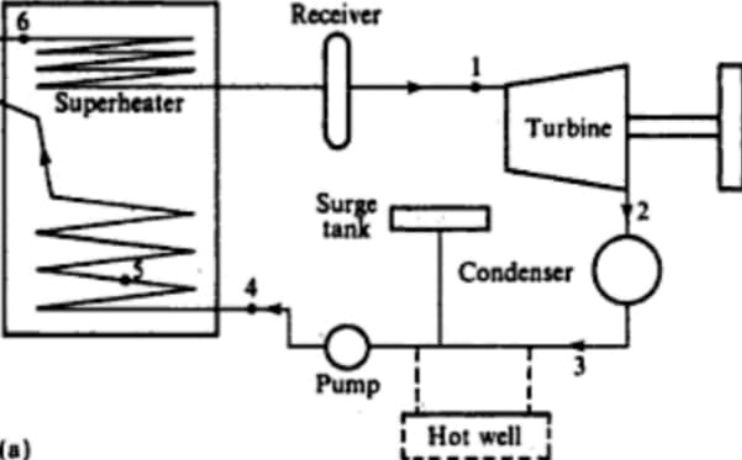
$$\begin{aligned} \text{Cycle efficiency} &= \frac{(h_1 - h_2) - (h_4 - h_3)}{\text{gross heat supplied}} \\ &= \frac{793.6 - 4.2}{(2800 - 112) - 4.2} = 0.294 \end{aligned}$$

i.e. $\text{Cycle efficiency} = 29.4\%$

$$\text{Work ratio} = \frac{-W_{12} - \text{pump work}}{-W_{12}} = \frac{793.6 - 4.2}{793.6} = 0.995$$

Also

$$\text{ssc} = \frac{1}{793.6 - 4.2} = 0.001267 \text{ kg/kW s} = 4.56 \text{ kg/kW h}$$



Example 8.2 Compare the Rankine cycle performance of Example 8.1 with that obtained when the steam is superheated to 500°C. Neglect the feed-pump work.

Solution From tables, by interpolation, at 42 bar:

$$h_1 = 3442.6 \text{ kJ/kg} \quad \text{and} \quad s_1 = s_2 = 7.066 \text{ kJ/kg K}$$

Using equation (4.10)

$$s_2 = s_{f_2} + x_2 s_{fg_2} \quad \text{therefore} \quad 0.391 + x_2 8.13 = 7.066$$

$$\text{i.e.} \quad x_2 = 0.821$$

Using equation (2.2)

$$h_2 = h_{f_2} + x_2 h_{fg_2} = 112 + (0.821 \times 2438) = 2113 \text{ kJ/kg}$$

From tables:

$$h_3 = 112 \text{ kJ/kg}$$

Then, using equation (8.2)

$$-W_{12} = h_1 - h_2 = 3442.6 - 2113 = 1329.6 \text{ kJ/kg}$$

Neglecting the feed-pump term, we have

$$\text{Heat supplied} = h_1 - h_3 = 3442.6 - 112 = 3330.6 \text{ kJ/kg}$$

Using equation (8.9)

$$\text{Cycle efficiency} = \frac{h_1 - h_2}{h_1 - h_3} = \frac{1329.6}{3330.6} = 0.399 \text{ or } 39.9\%$$

Also, using equation (8.14)

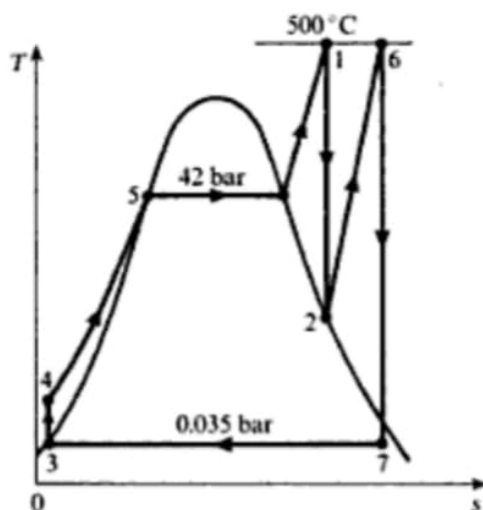
$$\text{ssc} = \frac{1}{h_1 - h_2} = \frac{1}{1329.6} = 0.000752 \text{ kg/kW s} = 2.71 \text{ kg/kW h}$$

Example 8.3

Calculate the new cycle efficiency and specific steam consumption if reheat is included in the plant of Example 8.2. The steam conditions at inlet to the turbine are 42 bar and 500 °C, and the condenser pressure is 0.035 bar as before. Assume that the steam is just dry saturated on leaving the first turbine, and is reheated to its initial temperature. Neglect the feed-pump term.

Solution The cycle is shown on a T - s diagram in Fig. 8.14.

Fig. 8.14 T - s diagram
Example 8.3

**Cycles**

It is convenient to read off the values of enthalpy from the h - s chart, i.e. $h_1 = 3442.6$ kJ/kg; $h_2 = 2713$ kJ/kg (at 2.3 bar); $h_6 = 3487$ kJ/kg (at 2.3 bar and 500 °C); $h_7 = 2535$ kJ/kg.

From tables

$$h_3 = 112 \text{ kJ/kg}$$

$$\begin{aligned} \text{Then Turbine work} &= (h_1 - h_2) + (h_6 - h_7) \\ &= (3443 - 2713) + (3487 - 2535) \end{aligned}$$

$$\text{i.e. Turbine work} = 1682 \text{ kJ/kg}$$

$$\begin{aligned} \text{Heat supplied} &= (h_1 - h_3) + (h_6 - h_2) \\ &= (3443 - 112) + (3487 - 2713) \end{aligned}$$

$$\text{i.e. Heat supplied} = 4105 \text{ kJ/kg}$$

therefore

$$\text{Cycle efficiency} = \frac{1682}{4105} = 0.41 \text{ or } 41\%$$

$$\text{Also } \text{ssc} = \frac{1}{1682} = 0.000595 \text{ kg/kJ}$$

$$\text{i.e. } \text{ssc} = 2.14 \text{ kg/kW h}$$

Q.2. (a) Show that the thermal efficiency of the dual combustion cycle is given by:-

$$\eta_{th} = 1 - \frac{1}{r_v^{(\gamma-1)}} \left\{ \frac{\alpha\beta^\gamma - 1}{(\alpha - 1) + \alpha\gamma(\beta - 1)} \right\}$$

Where

$r_v = v_1 / v_2$ = the adiabatic compression, volume ratio

$\beta = v_4 / v_3$ = cut-off ratio for constant pressure heat addition

$\alpha = p_3 / p_2$ = constant volume heat addition pressure ratio

(5 mrks)

- (b) An ideal dual combustion cycle has a volume ratio of the adiabatic compression of 15:1. At the beginning of the adiabatic compression the pressure, volume and temperature of the gas are 97 kN/m^2 , 0.084 m^3 and 28°C respectively. The maximum pressure and temperature of the cycle are 6.2 MN/m^2 and 1320°C , respectively.

For the cycle determine:

- The pressure, volume and temperature at the cycle process change points.
- The net work done
- The thermal efficiency
- The heat received
- The work ratio
- The mean effective pressure
- The carnot efficiency within the cycle temperature limits.

Take $C_p = 1.005 \text{ kJ/kgK}$, $C_v = 0.717 \text{ kJ/kgK}$

(15 mrks)

Q.3 The cycle of operations in a compression-ignition engine may be represented approximately as follows:-

- Compression, according to $pV^{1.3} = \text{constant}$, through a volume compression ratio of 14:1 from an initial pressure of 1 bar,
- Constant volume combustion to a maximum pressure of 50 bar,
- Constant pressure combustion during which the volume increases by 50%
- Expansion, according to $pV^{1.3} = \text{constant}$, to the original volume,
- Reduction of pressure at constant volume to the original pressure of 1 bar.

(a) Sketch a P-V diagram showing these operations

(3 mrks)

(b) Calculate the mean effective pressure

(9 mrks)

(c) Determine the cylinder bore and the stroke of such an engine operating on the four stroke cycle for a brake power of 1000KW at 3000rev/min if there are 4 cylinders and the piston stroke is equal to 1.25X cylinder bore diameter.

Assume the mechanical efficiency is 80%

(8 mrks)

Q.4 (a) Draw a labeled diagram of the Orsat apparatus used for combustion analysis of gases, stating the order in which the product gases are absorbed in the apparatus. (5 mrks)

(b) A petrol fuel has the following ultimate analysis:- C 83.7%; H16.3% The dry product analysis by volume is:- CO_2 11.8%; O_2 3.7%; N_2 84.5%

Determine, by Carbon-balance method, the air to fuel ratio used for combustion. (6 mrks)

- (c) (i) Determine the stoichiometric air fuel ratio for a petrol fuel approximating to hexane C_6H_{14} .
- (ii) Deduce the chemical equation if the petrol is burnt in 20 per cent excess air and also
- (iii) the wet volumetric analysis of the products if all the water vapour is present

(9 mrks)

Q.5. An open type gas turbine has two compressors in series. A high pressure turbine alone drives both compressors and a low-pressure turbine alone drives the external load. The pressure ratio is the same for each compressor, the overall pressure ratio for the engine being 1.6. The isentropic efficiency of each compressor is 0.85 and for each turbine 0.9. At inlet to the engine the air temperature and pressure are 20°C , 1 bar respectively. The air leaving the low-pressure compressor is cooled at constant pressure at 20°C before entering the high pressure compressor. The gas temperature at entry to the high pressure turbine is 800°C and on leaving this turbine the gas is reheated to 700°C before entering the low-pressure turbine. The exhaust from the low pressure turbine is used in a heat exchanger to raise the temperature of the air leaving the high pressure compressor by 75% of the maximum possible temperature rise.

- (a) Sketch (i) the layout of the plant (2 mrks)
(ii) a T-S diagram, (2 mrks)

Showing the processes involved and marking on both diagrams the corresponding state points.

(b) Determine:

- (i) The pressure of the gas leaving the high pressure turbine (4 mrks)
(ii) The specific work output (4 mrks)
(iii) The cycle efficiency and (4 mrks)
(iv) The cycle efficiency of a simple air-standard Joule cycle having the same overall pressure ratio. (4 mrks)

NB. Assume the compressor and turbines operate adiabatically and that pressure losses in the combustion chamber, heat exchanger and ducting are negligible. For compression $C_p = 1.01\text{ kJ/kg}\cdot\text{K}$ and $\gamma = 1.4$ For combustion and expansion $C_p = 1.15\text{ kJ/kg}\cdot\text{K}$ and $\gamma = 1.33$